

Experiment ‘p_4_4_month_with_highest_temperature_array’ Results

June 3, 2026

Experiment outcome: SUCCESS
Bad responses: 0
Responses containing assume : 0
Responses containing expect : 0
Resolution attempts: 1
Hard fails (resolution): 0
Soft fails (resolution): 0
Verification attempts: 1

Problem Specification

Problem name: p_4_4_month_with_highest_temperature_array

Natural language statement: Write a method that reads a sequence of twelve temperature values and returns the number of the month with the highest temperature.

Method signature: p_4_4_month_with_highest_temperature_array(inputs: array<real>) returns (hottest: int)

Ensures

- `1 <= hottest <= 12`
- `forall i :: 0 <= i <= inputs.Length - 1 ==> inputs[hottest - 1] >= inputs[i]`

Requires

- `inputs.Length == 12`

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method that reads a sequence of twelve temperature values and returns the number
↪ of the month with the highest temperature.

The signature should be:

p_4_4_month_with_highest_temperature_array(inputs: array<real>) returns (hottest: int)

The method should respect the following contract:

requires inputs.Length == 12, ensures 1 <= hottest <= 12, ensures forall i :: 0 <= i <=
↪ inputs.Length - 1 ==> inputs[hottest - 1] >= inputs[i]

Produce and show only the Dafny body of this method, including the curly braces that
↪ surround it. Do not show the signature nor contract. You must not use 'assume' nor
↪ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DmfXBekQvr0YD0tgd5fnCW5eVkt2k

Model: gpt-5.5-2026-04-23

Created at: 1780491773

Finish reason: stop

Usage: 354 tokens in, and 646 tokens out

```
{
  hottest := 1;
  var idx := 1;
  while idx < inputs.Length
    invariant 1 <= idx <= inputs.Length
    invariant 1 <= hottest <= idx
    invariant forall i :: 0 <= i < idx ==> inputs[hottest - 1] >= inputs[i]
    decreases inputs.Length - idx
  {
    if inputs[idx] > inputs[hottest - 1] {
      hottest := idx + 1;
    }
    idx := idx + 1;
  }
}
```

Final Program

```

method p_4_4_month_with_highest_temperature_array(inputs: array<real>) returns (hottest:
    ↪ int)
    requires inputs.Length == 12
    ensures 1 <= hottest <= 12
    ensures forall i :: 0 <= i <= inputs.Length - 1 ==> inputs[hottest - 1] >= inputs[
        ↪ i]
{
    hottest := 1;
    var idx := 1;
    while idx < inputs.Length
        invariant 1 <= idx <= inputs.Length
        invariant 1 <= hottest <= idx
        invariant forall i :: 0 <= i < idx ==> inputs[hottest - 1] >= inputs[i]
        decreases inputs.Length - idx
    {
        if inputs[idx] > inputs[hottest - 1] {
            hottest := idx + 1;
        }
        idx := idx + 1;
    }
}

```

Total Token Usage

Input tokens: 354

Output tokens: 646

Reasoning tokens: 512

Sum of 'total tokens': 1000

Experiment Timings

Overall Experiment started at 1780491773517, ended at 1780491793124, lasting 19607ms (19.61 seconds)

Iteration #1 started at 1780491773517, ended at 1780491793124, lasting 19607ms (19.61 seconds)